Solved Examples

Quadric Surfaces, Solids of Revolution, and Double Integrals

A. Quadric surfaces

1. Quick classification

Identify each surface and state its axis or opening direction.

$$\text{(a)} \quad x^2 + 4z^2 = 16,$$

$$\text{(b)} \quad z = x^2 + 4y^2,$$

$$\text{(c)} \quad \frac{z^2}{16} - \frac{x^2}{4} - \frac{y^2}{9} = 1.$$

Detailed solution

(a)

$$x^2 + 4z^2 = 16 \quad \Longleftrightarrow \quad \frac{x^2}{16} + \frac{z^2}{4} = 1.$$

The variable y is missing. Therefore the ellipse in the xz-plane is repeated for every value of y:

$$\boxed{\text{elliptic cylinder parallel to the } y\text{-axis.}}$$

(b)

$$z = x^2 + 4y^2.$$

The variable z is unsquared and the squared terms have the same sign. Hence

$$\boxed{\text{elliptic paraboloid opening in the positive } z\text{-direction.}}$$

(c)

$$\frac{z^2}{16} - \frac{x^2}{4} - \frac{y^2}{9} = 1.$$

Exactly one squared term is positive. Therefore

$$\boxed{\text{two-sheet hyperboloid with axis along the } z\text{-axis.}}$$

Key point

For an equation already written with right-hand side 1:

$$+ \ + \ - = 1$$

indicates a one-sheet hyperboloid, while

$$+ \ - \ - = 1$$

indicates a two-sheet hyperboloid. Confirm with a trace when in doubt.

2. Two-sheet hyperboloid: gap and traces

Identify and describe

$$\frac{z^2}{16} - \frac{x^2}{4} - \frac{y^2}{9} = 1.$$

Explain why it has two separate sheets and give two useful traces.

Detailed solution

There is exactly one positive squared term, $z^2/16$, so the surface is a two-sheet hyperboloid with axis along z. Rearrange:

$$\frac{z^2}{16} = 1 + \frac{x^2}{4} + \frac{y^2}{9}.$$

The right-hand side is at least 1, so

$$\frac{z^2}{16} \geq 1 \quad \Longrightarrow \quad |z| \geq 4.$$

Hence there are no points with $-4 < z < 4$. The two sheets begin at

$$\boxed{(0,0,4) \quad \text{and} \quad (0,0,-4)}.$$

For $y = 0$,

$$\frac{z^2}{16} - \frac{x^2}{4} = 1,$$

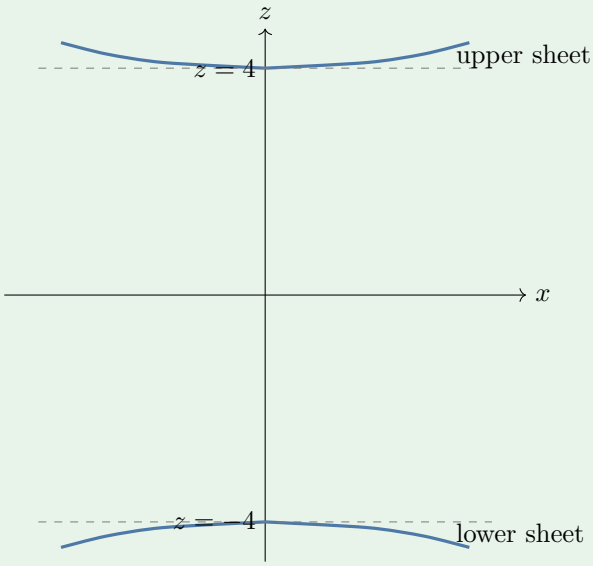
a hyperbola in the xz-plane.
For $z = 5$,

$$\frac{25}{16} - \frac{x^2}{4} - \frac{y^2}{9} = 1,$$

so

$$\frac{x^2}{4} + \frac{y^2}{9} = \frac{9}{16},$$

an ellipse.
Therefore the surface consists of elliptical cross-sections on two separated sides, $z \geq 4$ and $z \leq -4$.



B. Surface area and volume of revolution

3. Surface area about the y-axis

Find the surface area generated by rotating

$$y = x + 1, \qquad 0 \leq x \leq 2,$$

about the y-axis.

Detailed solution

For surface area,

$$S = \int 2\pi r \, ds.$$

Because the axis is the y-axis, the radius is $r = x$. Also,

$$\frac{dy}{dx} = 1,$$

so

$$ds = \sqrt{1 + 1^2}\, dx = \sqrt{2}\, dx.$$

Therefore

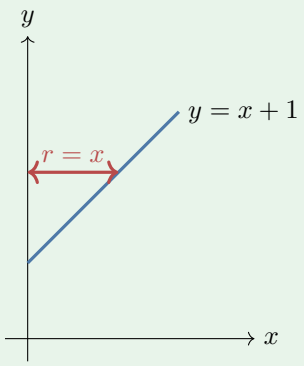
$$S = \int_0^2 2\pi x\sqrt{2}\,dx = 2\pi\sqrt{2}\int_0^2 x\,dx.$$

Thus

$$S = 2\pi\sqrt{2}\left[\frac{x^2}{2}\right]_0^2 = 4\pi\sqrt{2}.$$

Hence

$$\boxed{S = 4\pi\sqrt{2}}.$$



Key point

The formula is always

$$\boxed{S = \int 2\pi(\text{distance to axis})\,ds}.$$

Only the geometric radius changes with the axis of rotation.

4. Washer volume between $y = 2x$ and $y = x^2$

The region between

$$y = 2x \qquad \text{and} \qquad y = x^2$$

is rotated about the x-axis. Find the volume.

Detailed solution

The curves intersect when

$$2x = x^2 \quad \Longrightarrow \quad x(x-2) = 0,$$

so $x = 0, 2$.
For $0 < x < 2$, $2x > x^2$. Therefore

$$R(x) = 2x, \qquad r(x) = x^2.$$

The washer area is

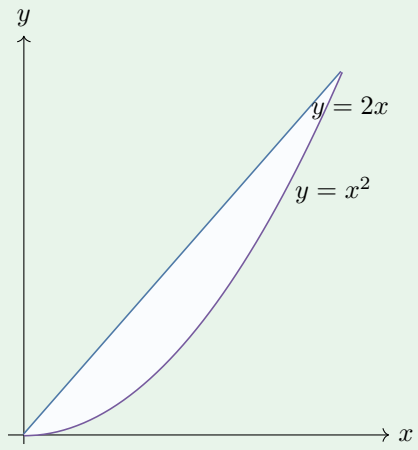
$$A(x) = \pi(R^2 - r^2) = \pi(4x^2 - x^4).$$

Hence

$$V = \pi\int_0^2 (4x^2 - x^4)\,dx.$$

Therefore

$$\begin{aligned} V &= \pi\left[\frac{4x^3}{3} - \frac{x^5}{5}\right]_0^2 \\ &= \pi\left(\frac{32}{3} - \frac{32}{5}\right) \\ &= \boxed{\frac{64\pi}{15}}. \end{aligned}$$



5. Washer volume between $y = 4$ and $y = x^2$

Find the volume obtained when the region bounded by

$$y = 4, \qquad y = x^2, \qquad -2 \le x \le 2$$

is rotated about the x-axis.

Detailed solution

The outer radius is

$$R = 4,$$

and the inner radius is

$$r = x^2.$$

Thus

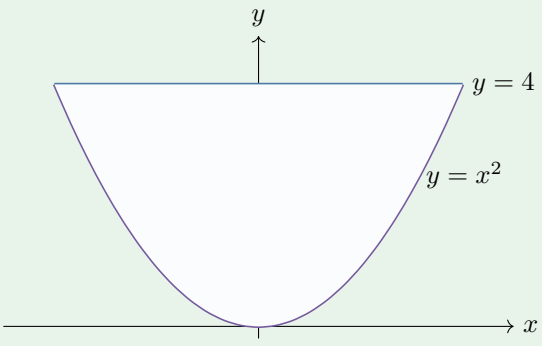
$$V = \pi \int_{-2}^{2} (16 - x^4)\, dx.$$

The integrand is even, so

$$V = 2\pi \int_{0}^{2} (16 - x^4)\, dx.$$

Hence

$$\begin{aligned} V &= 2\pi \left[16x - \frac{x^5}{5}\right]_0^2 \\ &= 2\pi \left(32 - \frac{32}{5}\right) \\ &= \boxed{\frac{256\pi}{5}}. \end{aligned}$$



Common mistake

For a washer,

$$A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2),$$

not $\pi(R - r)^2$.

C. Double integrals over general regions

Reading the bounds

For $dy\,dx$, hold x fixed and move vertically:

$$\boxed{\text{bottom } y\text{-boundary} \longrightarrow \text{top } y\text{-boundary.}}$$

For $dx\,dy$, hold y fixed and move horizontally:

$$\boxed{\text{left } x\text{-boundary} \longrightarrow \text{right } x\text{-boundary.}}$$

6. Type-I region bounded by $y = x^3$ and $y = x$

Let R be the first-quadrant region bounded by

$$y = x^3 \qquad \text{and} \qquad y = x.$$

Evaluate

$$\iint_R (2x + y)\,dA.$$

Detailed solution

The intersections satisfy

$$x^3 = x \quad \Longrightarrow \quad x(x^2 - 1) = 0.$$

In the first quadrant the relevant values are $x = 0, 1$.
For $0 < x < 1$, $x^3 < x$. Hence

$$0 \le x \le 1, \qquad x^3 \le y \le x.$$

Therefore

$$\iint_R (2x + y)\,dA = \int_0^1 \int_{x^3}^{x} (2x + y)\,dy\,dx.$$

Now

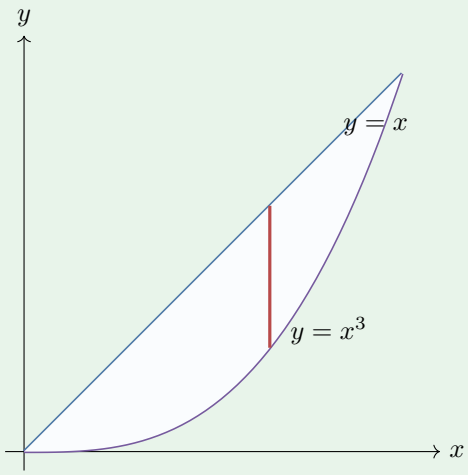
$$\int (2x + y)\,dy = 2xy + \frac{y^2}{2}.$$

Thus

$$\begin{aligned}\int_{x^3}^{x} (2x + y)\,dy &= \left[2xy + \frac{y^2}{2}\right]_{x^3}^{x} \\ &= \frac{5}{2}x^2 - 2x^4 - \frac{1}{2}x^6.\end{aligned}$$

Hence

$$\begin{aligned}\iint_R (2x + y)\,dA &= \int_0^1 \left(\frac{5}{2}x^2 - 2x^4 - \frac{1}{2}x^6\right) dx \\ &= \left[\frac{5}{6}x^3 - \frac{2}{5}x^5 - \frac{1}{14}x^7\right]_0^1 \\ &= \frac{5}{6} - \frac{2}{5} - \frac{1}{14} \\ &= \boxed{\frac{38}{105}}.\end{aligned}$$



7. Type-II region bounded by $x = y^2$ and $x = 2 - y$

Let R be bounded by

$$x = y^2 \qquad \text{and} \qquad x = 2 - y.$$

Evaluate

$$\iint_R (x + 1)\, dA$$

using horizontal slices.

Detailed solution

The intersection values satisfy

$$y^2 = 2 - y \quad \Longrightarrow \quad y^2 + y - 2 = 0.$$

Thus

$$(y + 2)(y - 1) = 0,$$

so

$$-2 \le y \le 1.$$

At $y = 0$, the two curves give $x = 0$ and $x = 2$. Therefore $x = y^2$ is the left boundary and $x = 2 - y$ is the right boundary.
Hence

$$\iint_R (x + 1)\, dA = \int_{-2}^{1} \int_{y^2}^{2-y} (x + 1)\, dx\, dy.$$

Since

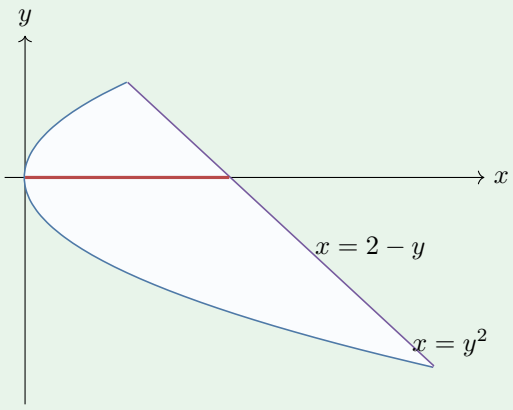
$$\int (x + 1)\, dx = \frac{x^2}{2} + x,$$

we get

$$\begin{aligned} \int_{y^2}^{2-y} (x + 1)\, dx &= \left[\frac{x^2}{2} + x\right]_{y^2}^{2-y} \\ &= 4 - 3y - \frac{1}{2}y^2 - \frac{1}{2}y^4. \end{aligned}$$

Therefore

$$\begin{aligned} \iint_R (x + 1)\, dA &= \int_{-2}^{1} \left(4 - 3y - \frac{1}{2}y^2 - \frac{1}{2}y^4\right) dy \\ &= \left[4y - \frac{3}{2}y^2 - \frac{1}{6}y^3 - \frac{1}{10}y^5\right]_{-2}^{1} \\ &= \boxed{\frac{117}{10}}. \end{aligned}$$



8. Area of a region involving $|x|$

Find the area of

$$R = \{(x, y) : -2 \le x \le 2,\ |x| \le y \le 2\}$$

using a double integral.

Detailed solution

Area is obtained by integrating 1:

$$A = \iint_R 1 \, dA.$$

Thus

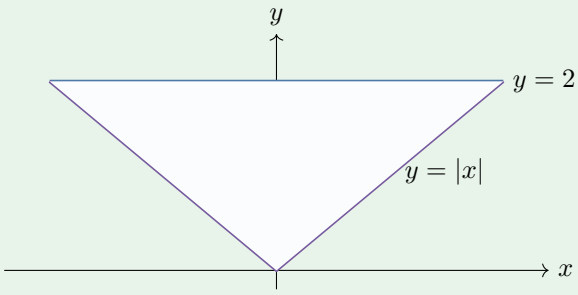
$$A = \int_{-2}^{2} \int_{|x|}^{2} 1 \, dy \, dx = \int_{-2}^{2} (2 - |x|) \, dx.$$

The integrand is even, so

$$A = 2 \int_{0}^{2} (2 - x) \, dx.$$

Therefore

$$A = 2 \left[2x - \frac{x^2}{2} \right]_0^2 = 2(4 - 2) = \boxed{4}.$$



Check

The region is a triangle of base 4 and height 2, so $\frac{1}{2}(4)(2) = 4$, confirming the integral.

9. Region bounded by $y = x^2$ and $y = 2x$

Let R be bounded by

$$y = x^2 \qquad \text{and} \qquad y = 2x.$$

Evaluate

$$\iint_R (x + y) \, dA.$$

Detailed solution

The curves intersect when

$$x^2 = 2x \quad \Longrightarrow \quad x = 0, 2.$$

For $0 < x < 2$, $x^2 < 2x$. Therefore

$$0 \le x \le 2, \qquad x^2 \le y \le 2x.$$

Hence

$$\iint_R (x+y)\,dA = \int_0^2 \int_{x^2}^{2x} (x+y)\,dy\,dx.$$

Now

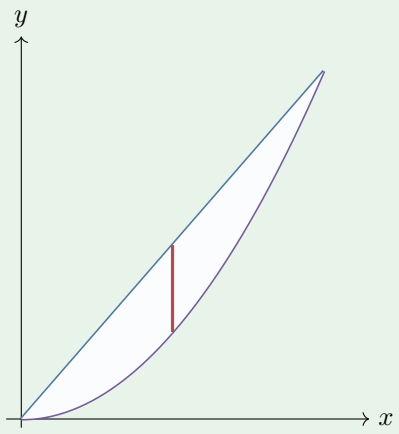
$$\int (x+y)\,dy = xy + \frac{y^2}{2}.$$

So

$$\begin{aligned}\int_{x^2}^{2x} (x+y)\,dy &= \left[xy + \frac{y^2}{2}\right]_{x^2}^{2x} \\ &= 4x^2 - x^3 - \frac{x^4}{2}.\end{aligned}$$

Therefore

$$\begin{aligned}\iint_R (x+y)\,dA &= \int_0^2 \left(4x^2 - x^3 - \frac{x^4}{2}\right) dx \\ &= \left[\frac{4x^3}{3} - \frac{x^4}{4} - \frac{x^5}{10}\right]_0^2 \\ &= \frac{32}{3} - 4 - \frac{16}{5} \\ &= \boxed{\frac{52}{15}}.\end{aligned}$$



10. Area enclosed by $y = x$ and $y = x^2$

Find the area enclosed by

$$y = x \qquad \text{and} \qquad y = x^2$$

using a double integral.

Detailed solution

The curves meet when $x = x^2$, so $x = 0, 1$. For $0 \le x \le 1$, $x^2 \le x$. Therefore

$$A = \int_0^1 \int_{x^2}^{x} 1\,dy\,dx.$$

Hence

$$A = \int_0^1 (x - x^2)\,dx = \left[\frac{x^2}{2} - \frac{x^3}{3}\right]_0^1 = \boxed{\frac{1}{6}}.$$

D. Double integrals in polar coordinates

Polar conversion

$$\boxed{x = r\cos\theta, \qquad y = r\sin\theta, \qquad x^2 + y^2 = r^2.}$$

Most importantly,

$$\boxed{dA = r\,dr\,d\theta.}$$

A small polar area has approximate side lengths dr and $r\,d\theta$.

11. Semicircular annulus with radial density

A thin lamina occupies

$$D = \{(x, y) : 1 \le x^2 + y^2 \le 9,\ y \ge 0\}.$$

Its areal density is

$$\rho(x, y) = \sqrt{x^2 + y^2}\ \text{kg/m}^2.$$

(a) Find the total mass. **(b)** Find the area of the same region using a polar double integral.

Detailed solution

The region is the upper half of the annulus between radii 1 and 3. Thus

$$1 \le r \le 3, \qquad 0 \le \theta \le \pi.$$

Also,

$$\rho = r, \qquad dA = r\,dr\,d\theta.$$

(a) Mass

The mass element is

$$dM = \rho\,dA = r^2\,dr\,d\theta.$$

Hence

$$M = \int_0^{\pi}\int_1^3 r^2\,dr\,d\theta.$$

Now

$$\int_1^3 r^2\,dr = \left[\frac{r^3}{3}\right]_1^3 = \frac{26}{3}.$$

Therefore

$$M = \int_0^{\pi}\frac{26}{3}\,d\theta = \boxed{\frac{26\pi}{3}\ \text{kg}}.$$

(b) Area

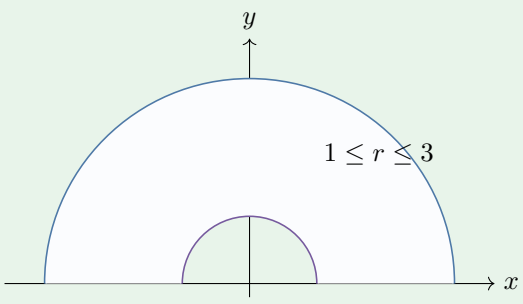
$$A = \iint_D 1\,dA = \int_0^{\pi}\int_1^3 r\,dr\,d\theta.$$

The inner integral is

$$\left[\frac{r^2}{2}\right]_1^3 = 4.$$

Thus

$$A = \int_0^{\pi} 4\,d\theta = \boxed{4\pi}.$$



12. Shifted circle converted to polar coordinates

Set up and evaluate, in polar coordinates, the area inside

$$x^2 + y^2 = 2x.$$

Detailed solution

Use $x^2 + y^2 = r^2$ and $x = r\cos\theta$:

$$r^2 = 2r\cos\theta.$$

Away from the origin,

$$r = 2\cos\theta.$$

The region occurs where $\cos\theta \geq 0$, so

$$-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}.$$

For each angle,

$$0 \leq r \leq 2\cos\theta.$$

Therefore

$$A = \int_{-\pi/2}^{\pi/2} \int_0^{2\cos\theta} r\, dr\, d\theta.$$

Integrating with respect to r,

$$A = 2\int_{-\pi/2}^{\pi/2} \cos^2\theta\, d\theta.$$

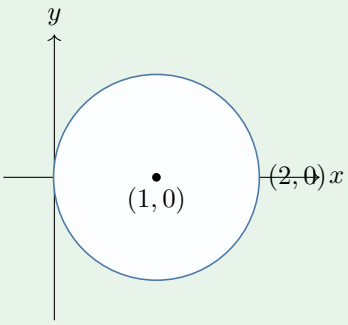
Using $\cos^2\theta = (1 + \cos 2\theta)/2$,

$$\boxed{A = \pi}.$$

As a check,

$$x^2 - 2x + y^2 = 0 \quad \Longrightarrow \quad (x-1)^2 + y^2 = 1,$$

which is a circle of radius 1.



13. Integral over a quarter-disk

Let

$$D = \{(x, y) : x^2 + y^2 \leq 4,\ x \geq 0,\ y \geq 0\}.$$

Evaluate

$$\iint_D (x^2 + y^2)\, dA.$$

Detailed solution

The region is a quarter-disk of radius 2:

$$0 \leq r \leq 2, \qquad 0 \leq \theta \leq \frac{\pi}{2}.$$

Also,

$$x^2 + y^2 = r^2, \qquad dA = r\, dr\, d\theta.$$

Hence

$$\iint_D (x^2 + y^2)\, dA = \int_0^{\pi/2} \int_0^2 r^3\, dr\, d\theta.$$

The inner integral is

$$\left[\frac{r^4}{4}\right]_0^2 = 4.$$

Therefore

$$\iint_D (x^2+y^2)\,dA = \int_0^{\pi/2} 4\,d\theta = \boxed{2\pi}.$$

14. Area of a disk using polar coordinates

Use a double integral in polar coordinates to find the area of

$$x^2+y^2 \le 9.$$

Detailed solution

The disk has radius 3, so

$$0 \le r \le 3, \qquad 0 \le \theta \le 2\pi.$$

Thus

$$A = \int_0^{2\pi}\int_0^3 r\,dr\,d\theta.$$

The inner integral is

$$\left[\frac{r^2}{2}\right]_0^3 = \frac{9}{2}.$$

Hence

$$A = \int_0^{2\pi} \frac{9}{2}\,d\theta = \frac{9}{2}(2\pi) = \boxed{9\pi}.$$